

User Documentation

[Jump to bottom](#)

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User Interface Documentation

This document describes the user interface of the Robot Visual Perception application. The interface is designed to provide real-time video streaming with advanced object detection and analysis capabilities.

Overview

The application consists of three main areas:

1. **Header:** Provides global controls, status indicators, and settings.
2. **Main View (Video Player):** Displays the real-time video stream with object detection overlays.
3. **Control Panel (Left Sidebar):** Contains detailed widgets for filtering objects, monitoring stream health, and viewing detection lists.

1. Header

The header bar at the top of the screen provides quick access to connection states and system preferences.

- **Panel Toggle:** A filter icon button on the left allows you to show or hide the Control Panel sidebar.
- **Language Selector:** A dropdown menu to switch the application language.
- **Application Title:** Displays the name of the application in the center.
- **Connection Controls:**
 - **Video Connection:** An icon button (Video/VideoOff) to connect or disconnect the raw video stream.
 - *Gray:* Disconnected.

- *Yellow*: Connecting.
 - *Green*: Connected.
 - **Analyzer Connection**: An activity pulse icon to connect or disconnect the analyzer service (metadata).
 - *Green*: Connected.
 - *Gray*: Disconnected.
 - **Theme Toggle**: Switches between Light and Dark mode.
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2. Main View (Video Player)

The central area displays the live camera feed.

Player Controls

Hovering over the video reveals the player controls at the bottom:

- **Play/Pause**: Toggles the video playback. Pausing allows you to inspect specific frames and clears the overlay to prevent clutter.
- **Fullscreen**: Expands the video to fill the entire screen.

Zoom Controls

You can zoom in/out of the video feed to inspect details:

- **Mouse Wheel**: Hold `Ctrl` (or `Meta` on Mac) and scroll to zoom in/out.

Interactive Overlay

When the Analyzer is connected, bounding boxes appear around detected objects.

- **Selection**: Clicking the "Select All" or specific class checkboxes in the Control Panel highlights only those objects.
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3. Control Panel (Left Sidebar)

The sidebar on the left organizes tools and information.

Connection & Actions

Located at the top of the panel.

- **Connection Buttons**: Mirrors the header controls for Video and Analyzer connections.

- **Clear Overlay:** Quickly clears all current selections and overlays.

Object Filter

Allows you to filter which detected objects are shown on the overlay.

- **Confidence Threshold:** A slider to set the minimum confidence level (0% - 100%). Detections below this confidence are hidden.
- **Action Buttons:**
 - **Select All:** Selects all currently detected classes.
 - **Clear All:** Deselects all classes.
- **Class List:** A dynamic list of detected object classes.
 - Each item shows the class name and the count of detections in the current frame.
 - Check/Uncheck a class to show/hide its bounding boxes.
- **Ordering:** New classes appear at the top of the list.

Stream Info

Displays technical statistics about the video stream and connection health.

- **Resolution:** Current dimensions of the video stream (e.g., 640 x 480).
- **Network Quality:**
 - **Packet Loss:** Percentage of data lost.
 - **Jitter:** Variance in packet arrival time (ms).
 - **Bitrate:** Current data transfer rate (Mbps).
- **Video Quality:** Counts of frames received and decoded.

Detections List

(Visible only when detections are present and Analyzer is connected)

- **Group By Type / Show Details:** A toggle button to switch between a grouped view (e.g., "3 Persons") or a detailed list of every individual object.
- **Content:** Lists detected objects with their specific details.

▼ Pages 5
Find a page or section...
▶ Home
▶ Build Documentation
▶ Fine Tuning

▸ **Technical Design Documentation**

▼ **User Documentation**

User Interface Documentation

Overview

1. Header

2. Main View (Video Player)

Player Controls

Zoom Controls

Interactive Overlay

3. Control Panel (Left Sidebar)

Connection & Actions

Object Filter

Stream Info

Detections List

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